Documented by:
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On behalf of:
True Interaction

LTB Sales Portal - System Development Progress

Third Milestone

The complexity and uncertainty of the data has made the second milestone somewhat challenging since we are still obtaining data that are essential for the analysis and development of the system. We are still performing analysis on the data as we obtain new sources or encounter uncertainties that can cause any complexity in the system; this is being done along with the implementation of the third milestone. The goals of this milestone has been successfully achieved, and all models have been created based on the data design and team discussions. Functions(controllers) of some models are currently missing because of their strong dependency to the user interface, and implementing these without the required interface makes testing very difficult or inaccurate if implemented. Below is a brief summary of the goals, and current progress, achieved to this point.

Milestone Goals:

The goals of this milestone were based on further analyzing the data, including new data sources, and developing the models and operation methods of the system.

1. The data analysis task was based on identifying the missing data sources, gathering and handling data, and detecting and correcting corrupt or inaccurate data from the provided data sources. A proper data preparation and cleaning cannot be performed until all necessary data is gathered.
   a. Data Preparation
   b. Data Evaluation

2. The second task, and main goal of this milestone, is the development of the system models. Initially the system was designed as an integration of two models, the permissions-based system and the strategy-plan system. However, the permissions-based system has been modified to meet client's requirements with a lower difficulty level. This was achieved by eliminating permissions validations from a database on each group, which could create a higher time complexity and cause anomalies in the system if not properly implemented. Also, integrating within groups the rights to certain operation methods that users can be assign to as needed. As a result, the system development has gone through the following phases:
   a. user-based system (permissions-based system)
      i. Groups & Users Models
   b. strategy-plan system
      i. Production (Finance) Models/Methods
      ii. Operation (Sales) Models/Methods

Project Progress:

A brief summary of the work done to achieve the goals mentioned above is being described below, to obtain a more descriptive information above the current state of the project you can access the files provided in the reference section of this report.

1. Explaining the inconsistency in the data would be too overwhelming for the main goal of this milestone. Nevertheless, you can have access to the latest identified and provided data sources through this link data handling. Also, you can have access to the latest datasets provided through this link Cliff Artifacts. Lastly, to keep up with some of the tests and validations that are currently being performed on the data, you can access the following link data analysis.

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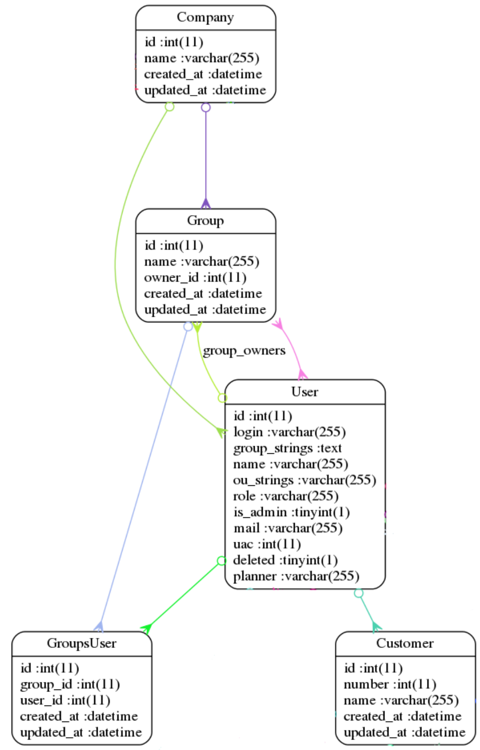
2. The user-based system is based on the user, customer and group models, whereas the strategy-plan system is based on the agreement, proforma, demand_solution, collection, event and item models. Any other model is considered dependent of the models mentioned above and, or, their integration.



User-Based System

User:

```
create_table "users", force: :cascade do |t|
  t.string  "login",         limit: 255
  t.text    "group_strings", limit: 65535
  t.string  "name",          limit: 255
  t.string  "ou_strings",    limit: 255
  t.string  "role",          limit: 255
  t.boolean "is_admin"
  t.string  "mail",          limit: 255
  t.integer "uac",           limit: 4
  t.boolean "deleted",                     default: false
  t.string  "planner",       limit: 255
end
```

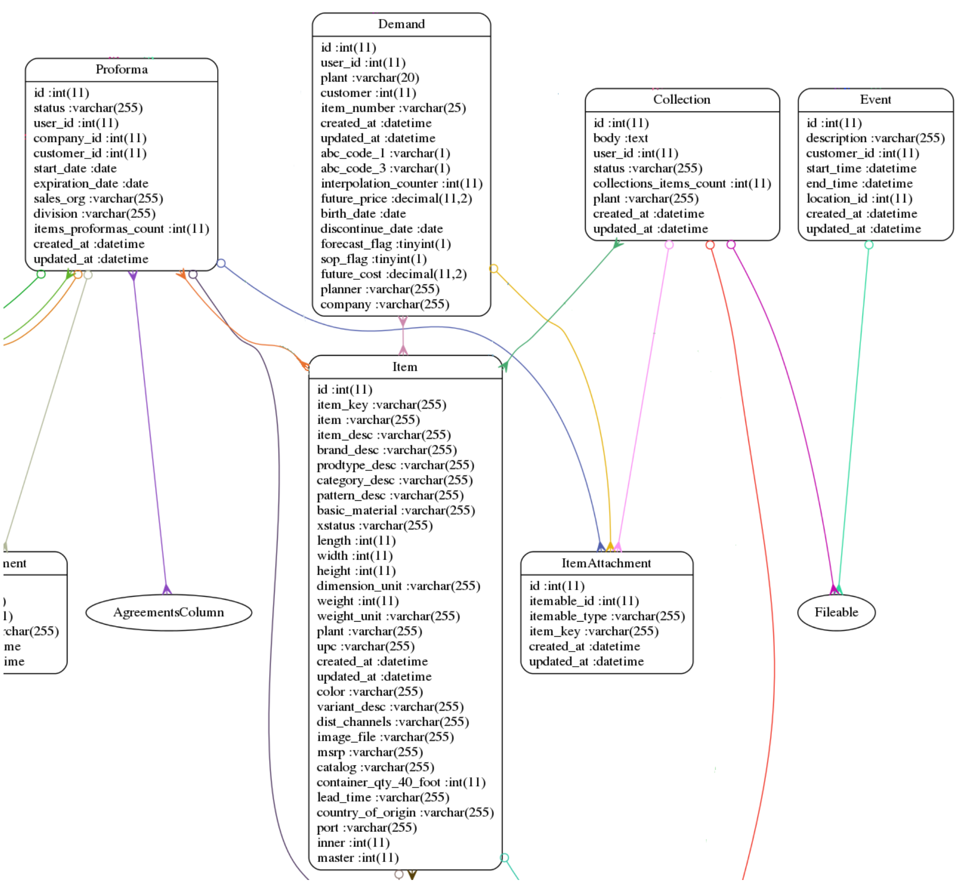
Group:

```
create_table "groups", force: :cascade do |t|
  t.string   "name",       limit: 255
  t.integer  "owner_id",   limit: 4
  t.datetime "created_at",             null: false
  t.datetime "updated_at",             null: false
end
```

Customer:

```
create_table "customers", force: :cascade do |t|
  t.integer  "number",     limit: 4
  t.string   "name",       limit: 255
  t.datetime "created_at",             null: false
  t.datetime "updated_at",             null: false
end
```

Strategy-Plan System



Collection:

```
create_table "collections", force: :cascade do |t|
  t.text     "body",                    limit: 65535
  t.integer  "user_id",                 limit: 4
  t.string   "status",                  limit: 255
  t.integer  "collections_items_count", limit: 4
  t.string   "plant",                   limit: 255
  t.datetime "created_at",              null: false
  t.datetime "updated_at",              null: false
end
```

Proforma:

```
create_table "proformas", force: :cascade do |t|
  t.string   "status",                limit: 255
  t.integer  "user_id",               limit: 4
  t.integer  "company_id",            limit: 4
  t.integer  "customer_id",           limit: 4
  t.date     "start_date"
  t.date     "expiration_date"
  t.string   "sales_org",             limit: 255
  t.string   "division",              limit: 255
  t.integer  "items_proformas_count", limit: 4
  t.datetime "created_at",            null: false
  t.datetime "updated_at",            null: false
end
```

Event:

```
create_table "events", force: :cascade do |t|
  t.string   "description", limit: 255
  t.integer  "customer_id", limit: 4
  t.datetime "start_time"
  t.datetime "end_time"
  t.integer  "location_id", limit: 4
  t.datetime "created_at",  null: false
  t.datetime "updated_at",  null: false
end
```

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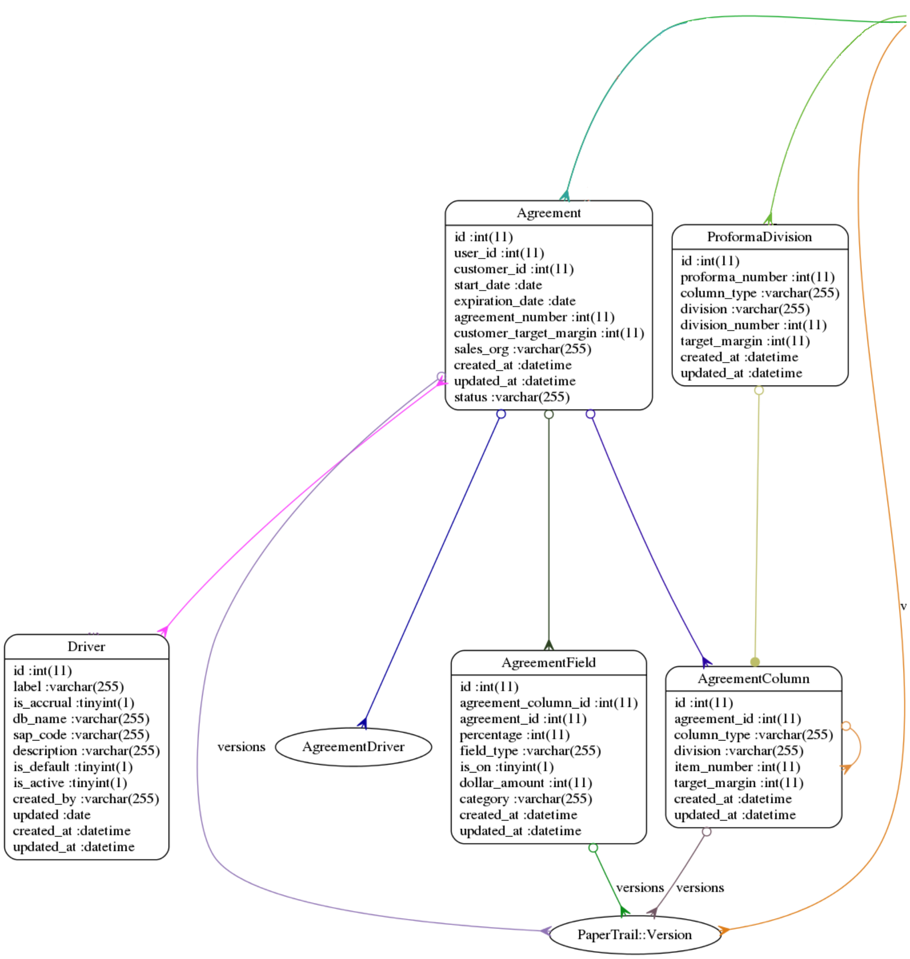
Demand:

```ruby
create_table "demands", force: :cascade do |t|
  t.integer  "user_id",                 limit: 4
  t.string   "plant",                   limit: 20
  t.integer  "customer",                limit: 4
  t.string   "item_number",             limit: 25
  t.datetime "created_at",
  t.datetime "updated_at",
  t.string   "abc_code_1",              limit: 1
  t.string   "abc_code_3",              limit: 1
  t.integer  "interpolation_counter",   limit: 4
  t.decimal  "future_price",            precision: 11, scale: 2
  t.date     "birth_date"
  t.date     "discontinue_date"
  t.boolean  "forecast_flag"
  t.boolean  "sop_flag"
  t.decimal  "future_cost",             precision: 11, scale: 2
  t.string   "planner",                 limit: 255
  t.string   "company",                 limit: 255
end
```

Item:

```ruby
create_table "items", force: :cascade do |t|
  t.string   "item_key",               limit: 255
  t.string   "item",                   limit: 255
  t.string   "item_desc",              limit: 255
  t.string   "brand_desc",             limit: 255
  t.string   "prodtype_desc",          limit: 255
  t.string   "category_desc",          limit: 255
  t.string   "pattern_desc",           limit: 255
  t.string   "basic_material",         limit: 255
  t.string   "xstatus",                limit: 255
  t.integer  "length",                 limit: 4
  t.integer  "width",                  limit: 4
  t.integer  "height",                 limit: 4
  t.string   "dimension_unit",         limit: 255
  t.integer  "weight",                 limit: 4
  t.string   "weight_unit",            limit: 255
  t.string   "plant",                  limit: 255
  t.string   "upc",                    limit: 255
  t.datetime "created_at",             null: false
  t.datetime "updated_at",             null: false
  t.string   "color",                  limit: 255
  t.string   "variant_desc",           limit: 255
  t.string   "dist_channels",          limit: 255
  t.string   "image_file",             limit: 255
  t.string   "msrp",                   limit: 255
  t.string   "catalog",                limit: 255
  t.integer  "container_qty_40_foot",  limit: 4
  t.string   "lead_time",              limit: 255
  t.string   "country_of_origin",      limit: 255
  t.string   "port",                   limit: 255
  t.integer  "inner",                  limit: 4
  t.integer  "master",                 limit: 4
end
```



Strategy-Plan System

Agreement:

```ruby
create_table "agreements", force: :cascade do |t|
  t.integer  "user_id",                  limit: 4
  t.integer  "customer_id",              limit: 4
  t.date     "start_date"
  t.date     "expiration_date"
  t.integer  "agreement_number",         limit: 4
  t.integer  "customer_target_margin",   limit: 4
  t.string   "sales_org",                limit: 255
  t.datetime "created_at",               null: false
  t.datetime "updated_at",               null: false
  t.string   "status",                   limit: 255
end
```

Driver:

```ruby
create_table "drivers", force: :cascade do |t|
  t.string   "label",        limit: 255
  t.boolean  "is_accrual"
  t.string   "db_name",      limit: 255
  t.string   "sap_code",     limit: 255
  t.string   "description",  limit: 255
  t.boolean  "is_default"
  t.boolean  "is_active"
  t.string   "created_by",   limit: 255
  t.date     "updated"
  t.datetime "created_at",   null: false
  t.datetime "updated_at",   null: false
end
```

The tables and diagrams above are only some samples of the current models that have been implemented in the system. Many tables have been omitted for the purpose of this report, however you can access all tables within the schema file of the system through this link schema. Also, you can access the MVC Application to have a deeper insight about the current development and any progress.

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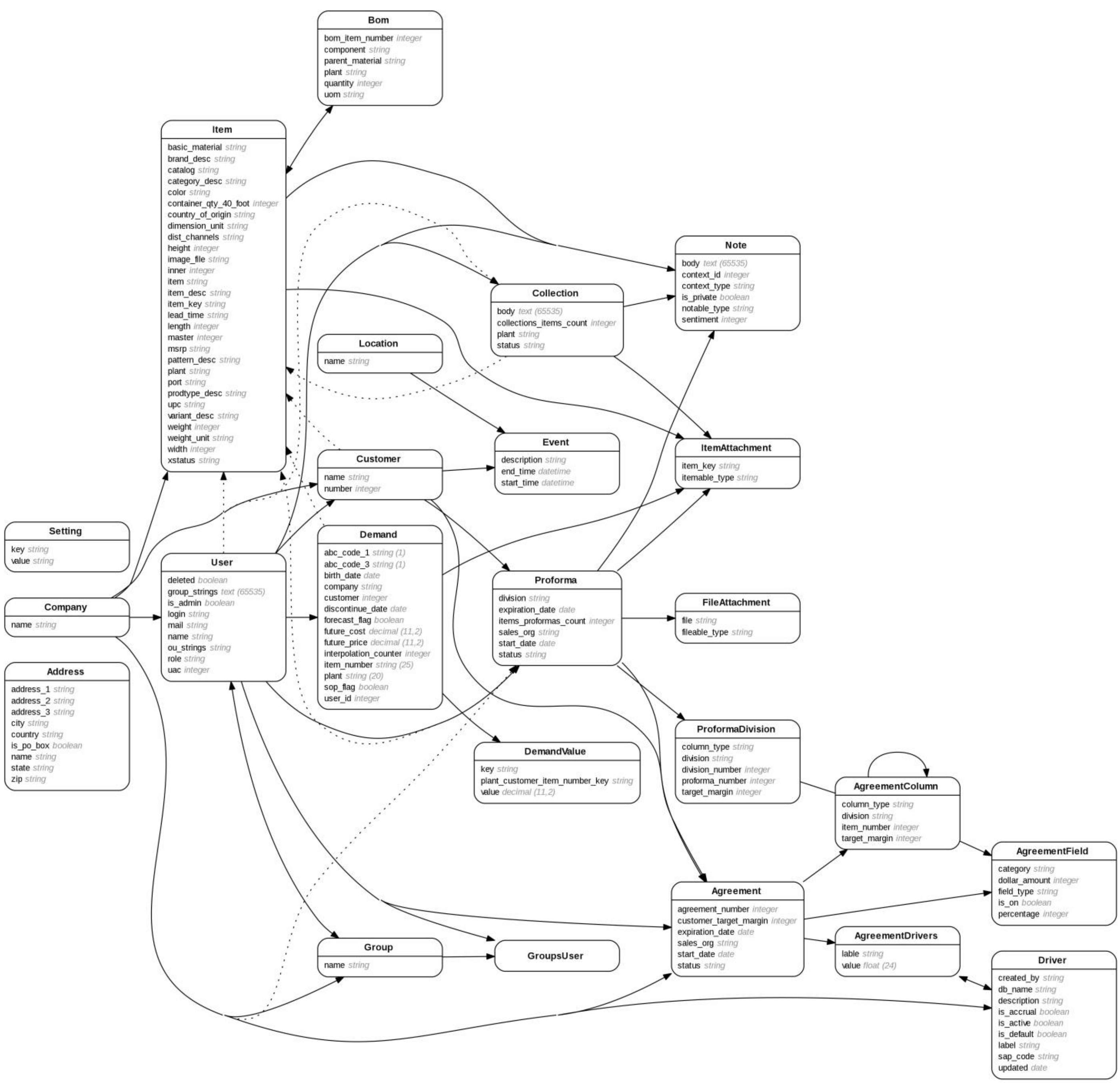
As illustrated above the development of the user-based and the strategy-plan systems have been implemented. The strategy-plan system contains the "Operation Methods" that are all methods utilized by sales group, which are also accessible by the finance group. This integration is the overall result of the "Life Time Brand(LTB) Sales Portal" system model development, which is being illustrated below.

Note: the current diagram of the system model development is subject to changes as its final review is scheduled at the end of this week. You can monitor any changes on the current system model development by accessing the [ERD report].

Salesportal domain model



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Next Step:
The next step, as briefly specified in the outline of this project, is the implementation of the models already created. This will involve the creation of controllers, those that are currently missing due to testing complexity, along with the graphics user interface to test and implement the system model as per design and meetings discussions. Once the user interface is done as per design, all models and methods must be tested to assure the system meets the client's requirements.

Reference Files:

System and Data Structure Design — **Week 0**

Sources

- Understanding the Company's Goal — LTB-Sales-Portal-Requested-Features
- Understanding the Data — Understanding-Data-Directory
- Design System Architecture — Data-Schema/PDFs/Model-Integration-System
- Design Data Structure — Wires/Frames-for-User-Interface

Data Analysis — **Week 4**

- Data Preparation — Data-Handling-Google-Spread-Sheet
- Data Exploration — Notes/Exploring-the-Data
- Modeling and Evaluation — Notes/Exploring-the-Data/Training-and-Testing

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System Development — **Week 8**

- Develop Permissions Based System Model — MVC Application
  - Users & Groups Methods
- Develop Strategic Plan System Model — schema
  - Finance Group Methods
  - Operations Methods
- Resulted System Model — ERD report